

CPSC 250 – Python for Data Manipulation

Test 2

Name: _____

Topics Covered: Functions, Parameter Passing, Classes and Objects, Constructors, Collections of Objects, and Operator Overloading

Total Points: 70

Part I – Multiple Choice (10 points)

Circle the best answer. Each question is worth 2 points.

1. What does the keyword `self` refer to inside a class?
 - A. The class itself
 - B. The current object
 - C. A constructor
 - D. A list of objects

2. What is the purpose of the `__init__()` method?
 - A. Print an object
 - B. Compare two objects
 - C. Initialize an object when it is created
 - D. Delete an object

3. What is the purpose of the `__str__()` method?
 - A. Create an object
 - B. Return a string representation of an object
 - C. Compare two objects
 - D. Modify an object

4. If a class defines an `__eq__()` method, which operator uses it?
 - A. +
 - B. *
 - C. ==
 - D. =

5. Which statement creates a new object named `book`?

- A. `book = Book`
- B. `Book = book()`
- C. `book = Book()`
- D. `create Book()`

Part II – Find the Error (15 points)

Each code segment contains one or more errors. Identify and correct them.

6.

```
class Student:
    def __init__(name):
        self.name = name
```

7.

```
class Student:
    def __init__(self, name):
        self.name = name

s = Student
print(s.name)
```

8.

```
students = []

students.append(Student)

print(students[0].name)
```

Part III – Code Writing (20 points)

9. Write a class called `Book` with the following:

- `title`
- `author`
- constructor, `__init__()`
- `__str__()` method

10. Given the `Book` class from Question 9, write code that:

- Creates three `Book` objects
- Stores them in a list called `library`
- Uses a loop to print all books

Part IV – Code Commentary (15 points)

Write comments describing what each section of code does.

```
# -----  
  
class Student:  
  
# -----  
  
    def __init__(self, name):  
        self.name = name  
  
# -----  
  
students = []  
  
# -----  
  
students.append(Student("Alice"))  
  
# -----  
  
for s in students:  
    print(s)
```

Part V – Short Answer (10 points)

11. Explain the difference between a class and an object.

12. Explain why a list of objects is often more useful than creating many separate variables such as

```
student1  
student2  
student3
```

Give at least one advantage.

End of Test